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A method for fast automated microscope image stitching

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ABSTRACT

Image stitching is an important technology to produce a panorama or larger image by combining several images with overlapped areas. In many biomedical researches, image stitching is highly desirable to acquire a panoramic image which represents large areas of certain structures or whole sections, while retaining microscopic resolution. In this study, we develop a fast normal light microscope image stitching algorithm based on feature extraction. At first, an algorithm of scale-space reconstruction of speeded-up robust features (SURF) was proposed to extract features from the images to be stitched with a short time and higher repeatability. Then, the histogram equalization (HE) method was employed to preprocess the images to enhance their contrast for extracting more features. Thirdly, the rough overlapping zones of the images preprocessed were calculated by phase correlation, and the improved SURF was used to extract the image features in the rough overlapping areas. Fourthly, the features were corresponded by matching algorithm and the transformation parameters were estimated, then the images were blended seamlessly. Finally, this procedure was applied to stitch normal light microscope images to verify its validity. Our experimental results demonstrate that the improved SURF algorithm is very robust to viewpoint, illumination, blur, rotation and zoom of the images and our method is able to stitch microscope images automatically with high precision and high speed. Also, the method proposed in this paper is applicable to registration and stitching of common images as well as stitching the microscope images in the field of virtual microscope for the purpose of observing, exchanging, saving, and establishing a database of microscope images.

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1. Introduction

In biomedical and clinical studies, it is often highly desirable to observe images of whole tissue sections with high resolution. However, it is usually difficult to get a single image with high resolution to view a whole sample. A usual solution to this problem is to stitch several images with overlapping areas to form a composite one (Rankov et al., 2005). Currently, there are many mosaic algorithms (Brown and Lowe, 2007) and softwares being applied in many fields. Some software, such as Autostitch, can realize automatic stitching of microscope images without any manual operation (Ma et al., 2007b), in which the main algorithm used for extracting features is Scale Invariant Feature Transform (SIFT) (Lowe, 2004). SIFT is robust to rotation, zoom, and illumination variation, but time consuming and a heavy computational burden. The objective of this paper is to develop a fast automated stitching algorithm to mosaic several microscope images to form a larger one. To realize the stitching of microscope images with high speed and high accuracy, SURF algorithm was chosen to extract image features in this paper, which has been proved to have a good performance than SIFT on feature extraction and can reduce the amount of computation (Bay et al., 2008).

In this paper, we improved SURF algorithm to get features of microscope images with a short time automatically. The stitching process can be divided into four steps: relative position estimation, features extraction, image registration, and image fusion. In the first step, we use phase correlation method (Kuglin and Hines, 1975; Balci and Foroosh, 2006) to acquire overlapping zones. During the features extraction, the improved SURF is used to extract features from the overlapping zones of two images to be stitched. After extraction, multi-images are compared to find the transformation parameters with Random sample consensus (RANSAC) (Fischler and Bolles, 1981) which can be used for the alignment of images. In the last step, the weighted average fusion strategy is used to eliminate blending seam to produce a seamless image. The algorithm developed in this paper is completely automated for fast stitching of microscope images.

2. Materials and methods

2.1. Animals

The animals used in our experiment are male adult mus musculus albus (25–35 g) obtained from Animal Institute of Xiangya

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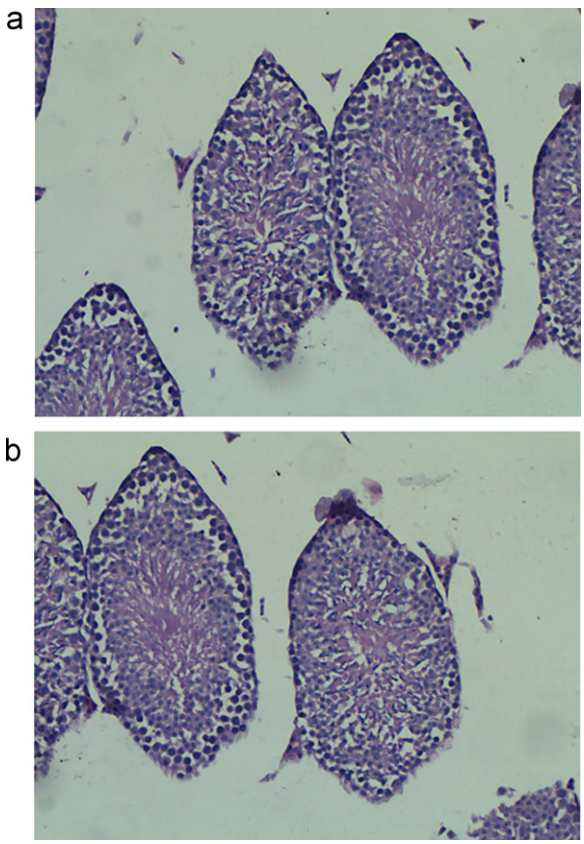

Fig. 1. Two images of rat's testicle obtained from microscope at different time with about 50% overlapping areas.

medical college. They were maintained in the animal laboratory of the Institute of Biomedical Engineering, School of Geosciences and Info-Physics, Central South University.

2.2. Tissue preparation

The rat was anesthetized and its testicle was quickly removed for experiment. The testicle histological section was dyed by hematoxylin and eosin staining. Fig. 1 shows the typical microscope images of rat's testicle.

2.3. Image acquisition

The imaging device is OLYMPUS CX31RTSF (OLYMPUS, Tokyo, Japan) normal light microscope, equipped with trinocular tube, a pair of eyepiece 10×, quadruple revolving nosepiece, mechanical-stage, abbe condenser, plan objective 4×, 10×, 40×, plan achromat objective 100×/1.24, motorized *x*, *y* stage, and a Cannon PowerShot A610 color camera.

40 groups of images were taken by 40× 0.75 NA objective lens with a resolution of 0.52 μm/pixels, and the image size is 768 × 576 pixels. These images were stored in the format of a BMP file stack.

2.4. Improvement of the SURF

2.4.1. The limitation of the SURF

According to SURF algorithm, when scales are more than 10, the features extracted are less (see Fig. 2). In order to obtain more features, it is important to increase the low scale and decrease the high scale. In addition, it is difficult to get the features matched correctly even though the number of extracted features is not small if there are great changes of viewpoint between the microscope images (see

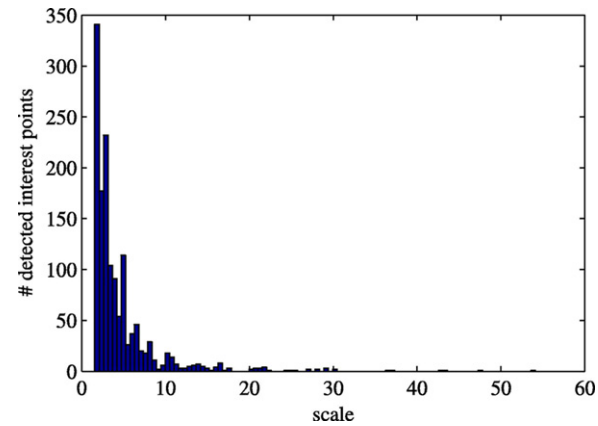

Fig. 2. Histogram calculation under different scales.

Fig. 3). It means that extracting features by the SURF algorithm is not robust to the changes of viewpoint. This problem can be solved by using affine transformations in advance to simulate the viewpoint changes of the images and by extracting the features from the images after affine transformations.

2.4.2. Derivation of the scale-space reconstruction of SURF

Because of the limitation of the SURF, the affine transformation is first applied to the images and then the scale spaces of SURF are reconstructed, to improve the robustness of the SURF algorithm to changes of viewpoint and to increase the number of the features extracted. The steps of feature detection with improved SURF algorithm are as follows:

(1) To simulate the various changes of viewpoint of the image, the transformation matrix A is taken here to apply affine transformation to the images (see expression 1), where ϕ is the angle by which the image rotates about the central axis, t is the gradient of the image and θ is the angle between the gradient image and the center normal, as is shown in Fig. 4.

$$A = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} t & 0 \\ 0 & 1 \end{bmatrix} \tag{1}$$

To simulate various changes of viewpoints as well as possible, the values of t and θ are given in Table 1. Where $t = 1/\cos\theta$, $\phi \in [0, \pi)$, $\theta \in [0, \pi/2)$, ϕ varies with θ, and the relationship is $\Delta\phi = 72^0/t$. In this manner, images with various viewpoints can be produced by the transformation matrix A. However, the computation amount will be very large in practice if features were extracted in all the images after affine transformation. Therefore, the images are firstly down sampled before affine transformation to reduce the size. The sampling factor K varies with the original size of the images, and it is defined as:

$$\frac{\text{image_size}}{K \times K} \geq 150 \tag{2}$$

To guarantee sufficient areas for extracting features, the image size after sampling should be not smaller than 150 × 150 pixel.

(2) Constructing a Hessian matrix using integral image and box-filters.

Table 1
Gradient t and angle θ.

t	1	$2\sqrt{3}/3$	$\sqrt{2}$	2	4
θ	0°	30°	45°	60°	75.5°

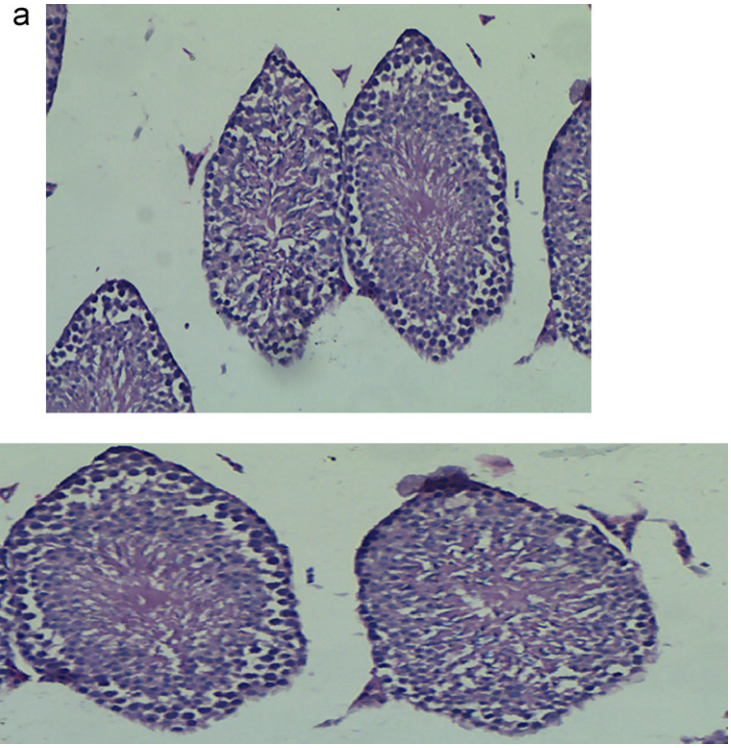

Fig. 3. The original microscope image and the one with viewpoint changed (b). (b) was acquired from (a) by formula (1).

Table 2
Scale-space reconstruction.

Octaves	Size of box-filter				
Octaves-1	9×9	15×15	21×21	27×27	33×33
Octaves-2	15×15	27×27	39×39	51×51	63×63

(3) Reconstructing scale spaces. The new scale spaces can be given by

$$fs = 3(2^i \times j + 1), \quad i = 1, 2, \quad j = 1, 2, 3, 4, 5 \tag{3}$$

where fs denote the size of box-filters, i is octave, j is layer. The whole ten box-filters are shown in Table 2.

(4) During the feature extraction, the method that how the features are extracted by SURF and the feature descriptors are kept without any change. The improvement is that the SURF algorithm is applied to extract the features and feature descriptors from the images after various affine transformations in step (1). The features and feature descriptors are saved to prepare for image registration. The whole process of the improvement is illustrated in Fig. 5.

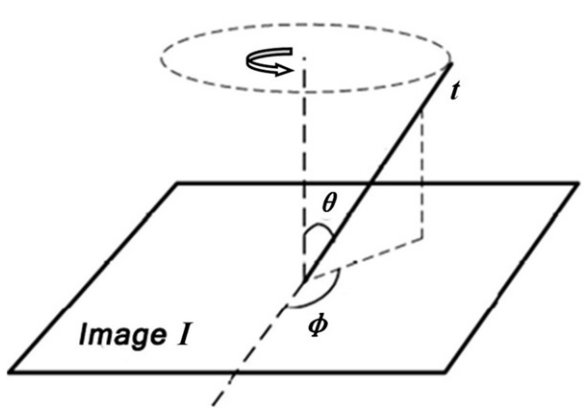

Fig. 4. Model of the transformation matrix A.

2.5. Stitching method for microscope images

2.5.1. Preprocessing

To extract features by SURF algorithm, the color images need converting to gray scale images. This process may lose some information in darker areas of the original images. Therefore, the gray scale image can be preprocessed by HE to enhance contrast. The gray scale and preprocessed images are shown in Fig. 6.

2.5.2. Relative position estimation by phase correlation

In this paper, phase correlation is used to estimate the rough relative position of the two images, original image and reference image, to be stitched. Let $f_1(x, y)$ and $f_2(x, y)$ denote original image and reference image, assuming there is a translation x_0 in x direction and a y_0 in y direction, we define

$$f_2(x, y) = f_1(x - x_0, y - y_0) \tag{4}$$

$$F_2(\xi, \eta) = F_1(\xi, \eta)e^{-j2\pi(\xi x_0 + \eta y_0)} \tag{5}$$

where $F_1(\xi, \eta)$ and $F_2(\xi, \eta)$ are the Fourier transformation of the two images, their cross-power spectrum is then given by

$$I(\xi, \eta) = \frac{F_1(\xi, \eta)F_2{}^*(\xi, \eta)}{\left|F_1(\xi, \eta)F_2{}^*(\xi, \eta)\right|} = e^{-j2\pi(\xi x_0 + \eta y_0)} \tag{6}$$

Therefore, we can find impulse function, $\delta(x - x_0, y - y_0)$, by inverse Fourier's transform to $I(\xi, \eta)$, the peak values of impulse function position can tell us the overlapping zones (As shown in Fig. 7).

Also if $f_1(x, y)$ and $f_2(x, y)$ have a rotation with θ_0 and minimization with σ, we can define

$$f_2(x, y) = f_1\left[\sigma^{-1}(x\cos\theta_0 + y\sin\theta_0), \quad \sigma^{-1}(-x\sin\theta_0 + y\cos\theta_0)\right] \tag{7}$$

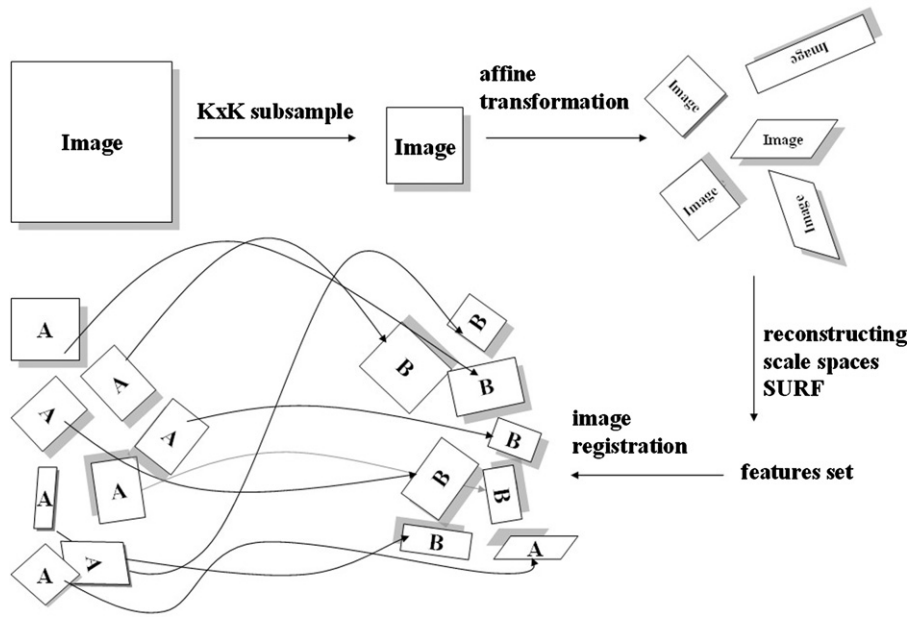

Fig. 5. The process of feature extraction and registration with improved SURF. A and B represent the two images with various affine transformations.

There we use $f_{1\rho}(\rho,\theta)$ and $f_{2\rho}(\rho,\theta)$ denote the polar coordinates of $f_1(x,y)$ and $f_2(x,y)$, so we can acquire

$$f_{2\rho}(\rho,\theta)=f_{1\rho}\left(\frac{\rho}{\sigma},\quad \theta-\theta_0\right) \tag{8}$$

And $f_{1\rho}(\rho,\theta)$ and $f_{2\rho}(\rho,\theta)$ are taken by the logarithm in ρ direction, so

$$f_{1\rho l}(\lambda,\theta)=f_{1\rho}(\log\rho,\theta),\quad f_{2\rho l}(\lambda,\theta)=f_{2\rho}(\log\rho,\theta) \tag{9}$$

$$f_{2\rho l}(\lambda,\theta)=f_{1\rho l}(\lambda-\lambda_0,\quad \theta-\theta_0) \tag{10}$$

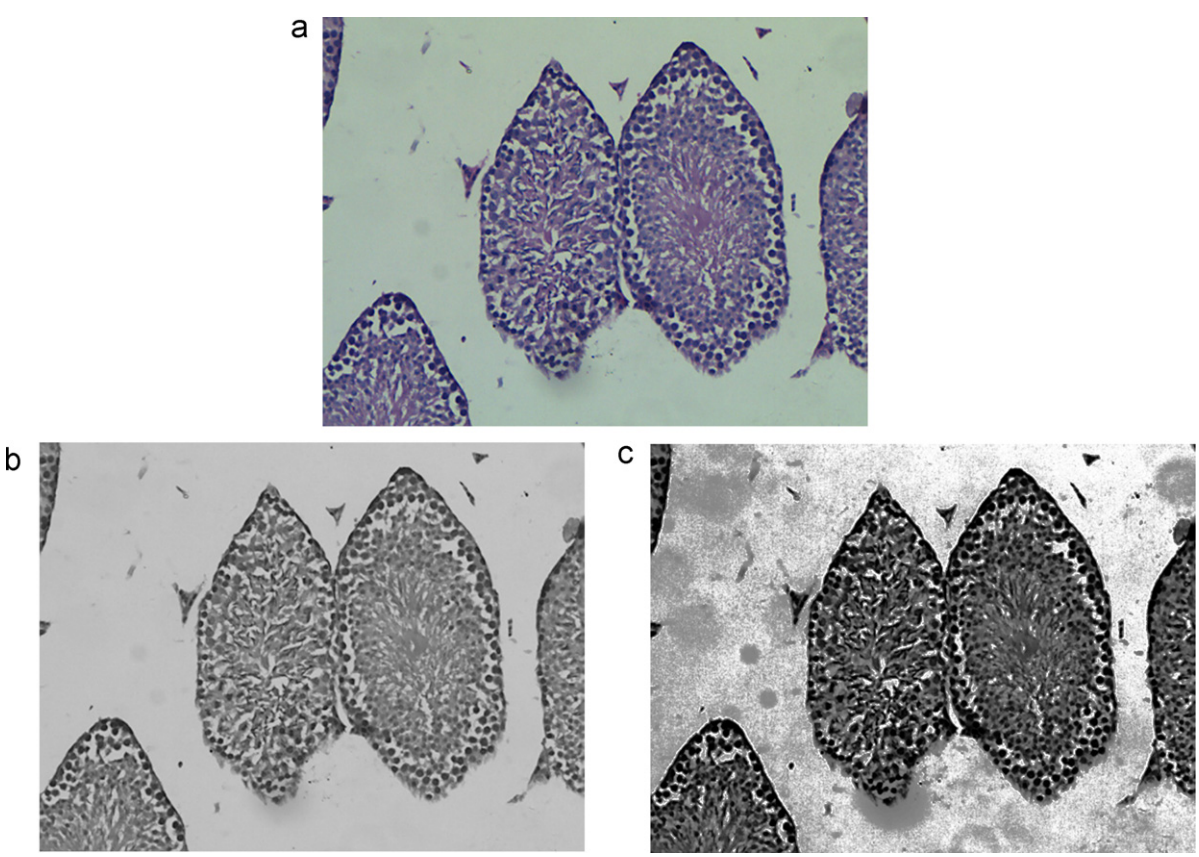

Fig. 6. (a) Original image; (b) gray scale image without histogram equalization; (c) gray scale image with histogram equalization.

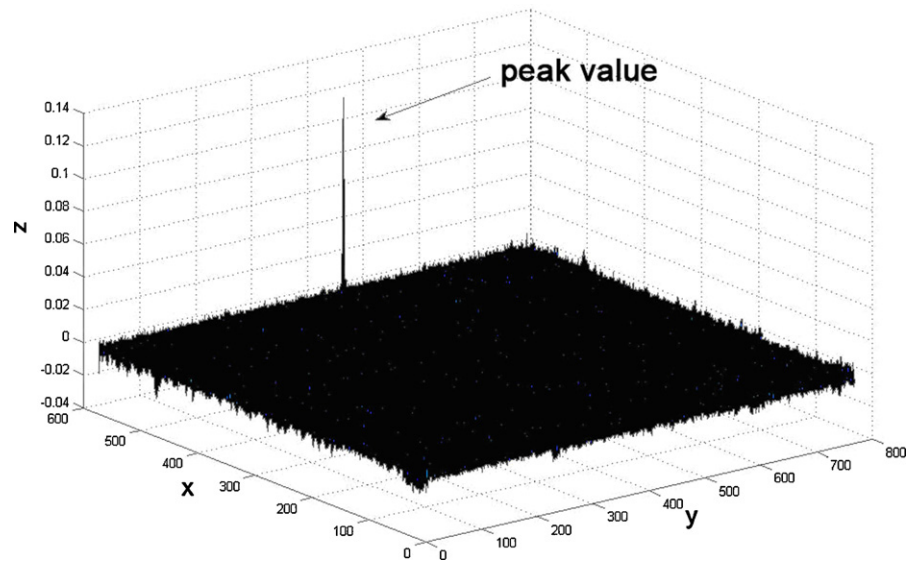

Fig. 7. Impulse function for two images.

where $\lambda_0 = \log\sigma$, from (4) and (10), we know $f_{1\rho l}$ and $f_{2\rho l}$ have a translation relationship, so we can compute the rotation θ_0 and minimization σ from (5) and (6).

2.5.3. Image registration

In the step mentioned above, we can find the peak value which stands for the overlapping zones in the two images to be stitched, but it is not an accurate position. So, special method must be employed to extract the features in the overlapping zones for precise matching. In this paper, we use the improved SURF developed ourselves to extract features. The matching method is to calculate Euclidean-Distance of features extracted from two images, and to find the nearest distance N1 and the second nearest distance N2. If N1 is less than 70% of N2, the point A in the original image matches point A' in the reference image. After having collected the location of each matching pairs, transformation parameters can be obtained via RANSAC. Hence, the accurate relative position is estimated.

2.5.4. Image fusion

In this step, weighted average fusion strategy is used to blend two images to eliminate blending seam. Let $f(x, y)$ denote the stitched image, each pixel value can be given by

$$f(x,y) = \begin{cases} f_1(x,y) & x,y \in f_1 \\ \omega_1 f_1(x,y) + \omega_2 f_2(x,y) & x,y \in (f_1 \cap f_2) \\ f_2(x,y) & x,y \in f_2 \end{cases} \tag{11}$$

$$\omega_1 = \frac{x - x_L}{x_R - x_L}, \quad \omega_2 = \frac{x_R - x}{x_R - x_L} \tag{12}$$

where x_L and x_R denote the left edge and right edge of overlapping zones of the two images. In this way, the seamless stitched image can be obtained.

3. Results

3.1. The performance of improved SURF

In order to evaluate the performance of improved SURF, we choose the images from the standard database of Mikolajczyk (Mikolajczyk, 2007) to test the robustness of this algorithm to rotation, zoom, blur, viewpoint, and illumination. There are 4 groups of images in Fig. 8. Each group consists of 6 images with different rotation and zoom, blur, viewpoint, or illumination. The image format is PPM.

For comparison, we choose the same threshold value for improved SURF and SURF. The first one of each group is chosen as the reference image. The correct matches of features detected by improved SURF and SURF are shown in Fig. 9, where the *y*-axis represents the number of correct matches and the *x*-axis represents the other 5 images in each group.

3.2. Comparison of feature detection with HE

improved SURF algorithms are applied to normal light microscope images with the pixel size of 768 × 576 at 40× magnification. We compared the algorithms with *and without HE*, and threshold value is 10,000. The result was shown in Fig. 10.

3.3. Comparison of matching results with HE

In this section, the images shown in Fig. 1(a) and (b) were used to compare the matching results of features detected by improved SURF with or without preprocessing images by HE. Matched corresponding features in the two images by improved SURF with or without HE are shown in Fig. 11. Table 3 shows the number of features detected and matched acquired by improved SURF with or without HE.

3.4. Comparison of matching performance with great changes of viewpoint

Here we use the microscope images shown in Fig. 3 to verify the matching performance of the improved SURF for the images with changes of viewpoint, and the results are shown in Fig. 12.

Table 3
Comparison between matching features with and without HE.

Algorithm	Number of features from Fig. 1(a)	Number of features from Fig. 1(b)	Matched feature pairs
Without HE	203	141	111
With HE	492	376	247

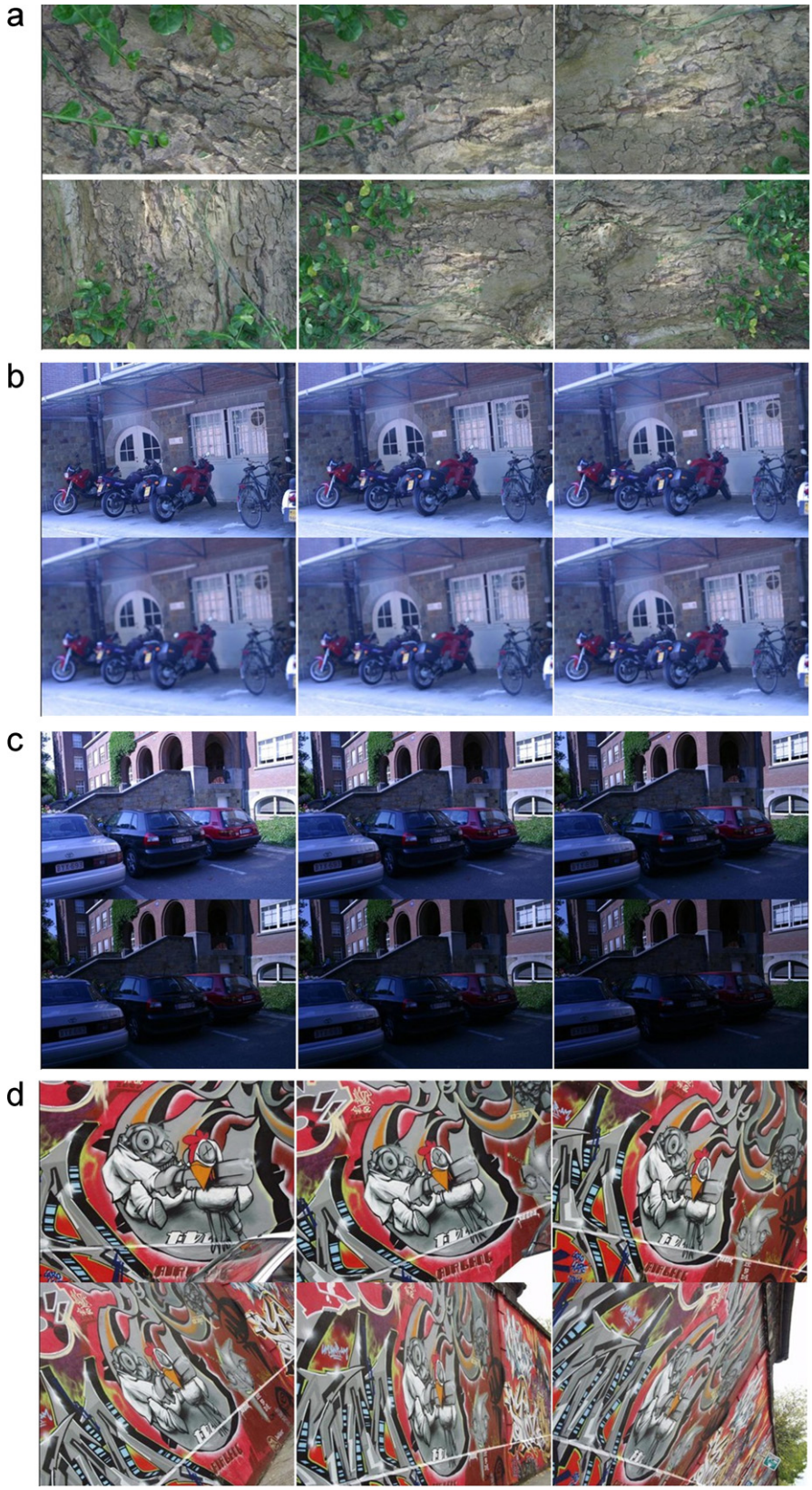

Fig. 8. 4 groups of images used for testing the improved SURF: (a) zoom + rotation, (b) blur, (c) illumination, (d) viewpoint.

3.5. Comparison of stitching performance

For evaluating the algorithm proposed in this study, 40 groups of microscope images obtained in different section at different time were stitched by this method and the software named Autostitch. The stitched results of one group (shown in Fig. 1) are shown in Fig. 13. Those two methods are compared under the same system environment, and the performance characters are listed in Table 4.

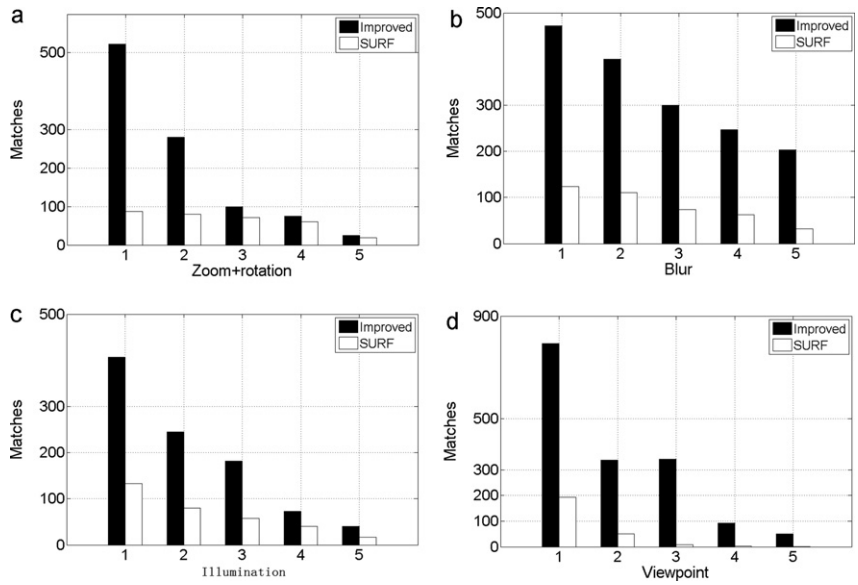

Fig. 9. Comparison between the correct matches of features detected by improved SURF and SURF.

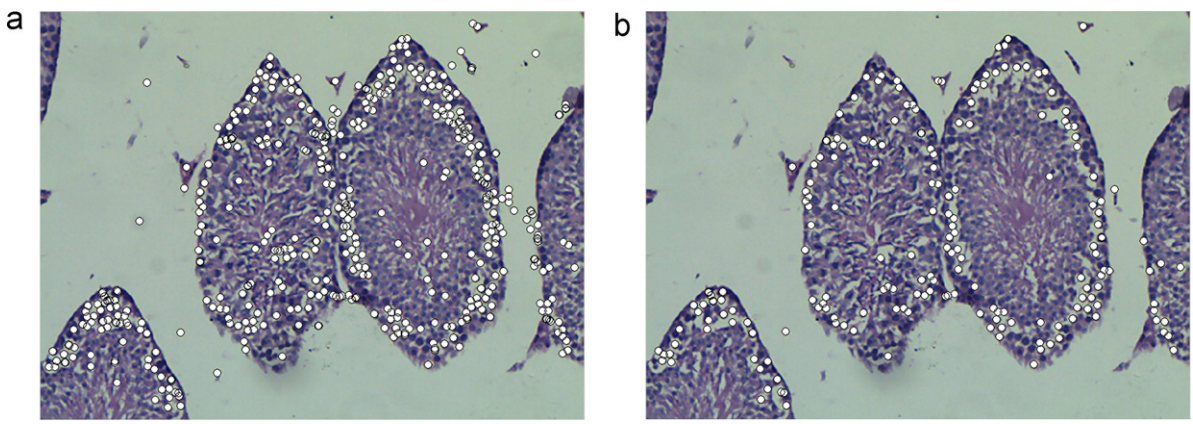

Fig. 10. (a) Features detected with HE; (b) features detected without HE. Note the white points denote features.

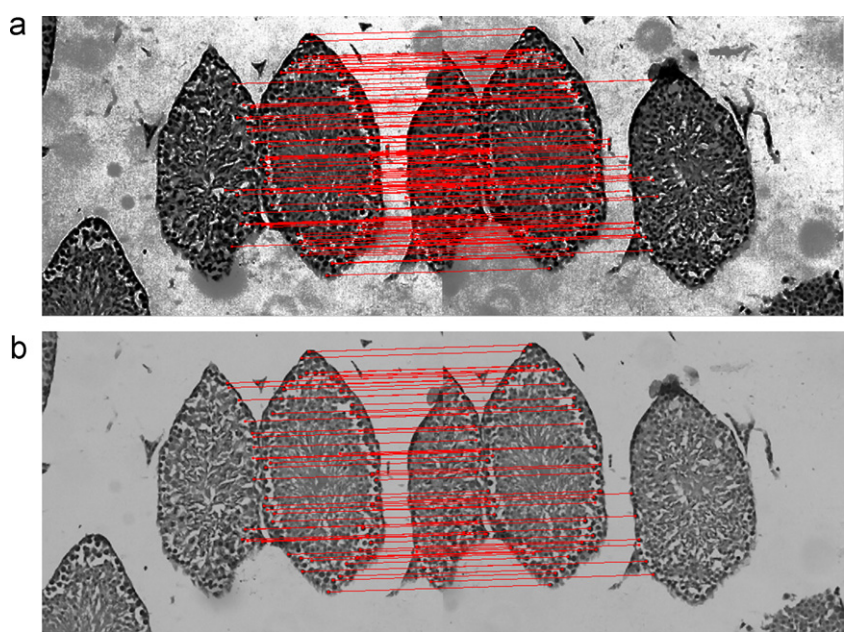

Fig. 11. (a) Matching features obtained with HE; (b) matching features obtained without HE.

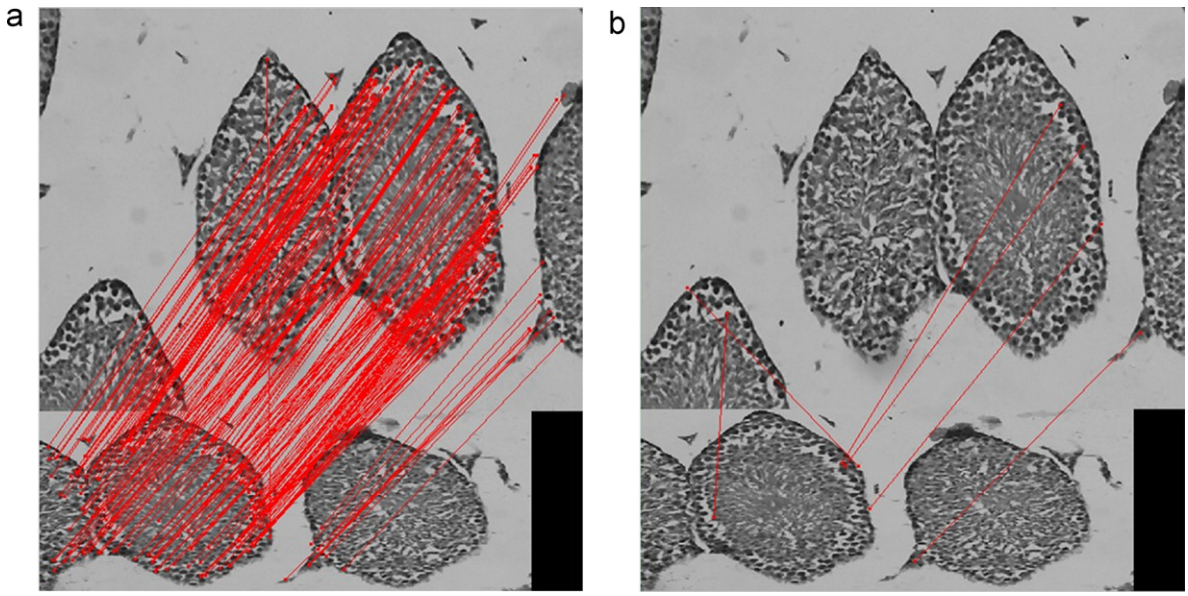

Fig. 12. The results of matching features detected by improved SURF (a) and SURF (b).

4. Discussion

4.1. Feature extraction by improved SURF and conventional SURF

In experiment 3.1, the SURF and Improved SURF are tested for the rotation and zoom, blur, illumination, and changes of viewpoint. Fig. 9 shows that the improved SURF is robust to all of the four kinds of changes and the matched features are much more than that of SURF. A relative large number of matched features can be obtained by improved SURF even if there are great changes of viewpoint. In experiment 3.4, the matching performance of improved SURF has been verified by microscope images (see Fig. 12).

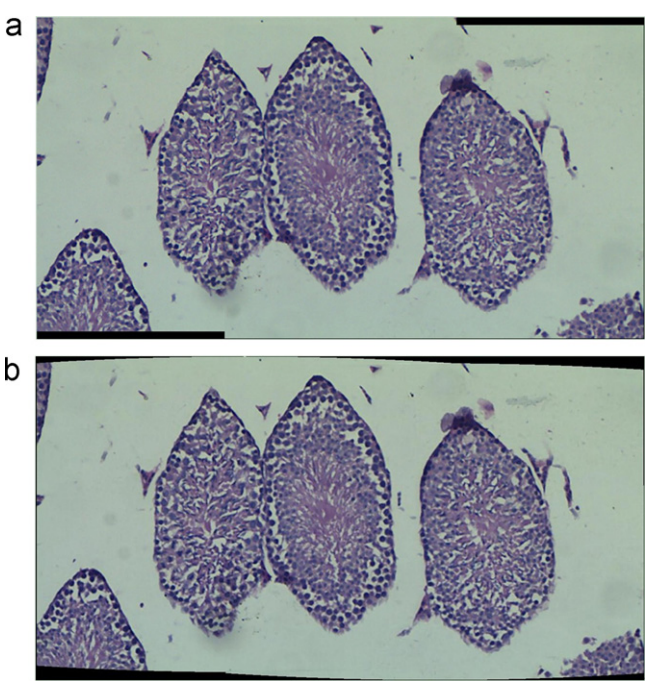

Fig. 13. Stitched results of two microscope images by the proposed method (a) and Autostitch(b).

4.2. Feature extraction by improved SURF with HE

In Fig. 11, the number of detected features with HE is much larger than those without HE in the gray images. It is known that as the number of the detected features increases, the performance of the recognition and matching process are enhanced. Therefore, it is noted from Fig. 11(a) that improved SURF with HE performs better with respect to the number of the features detected. Although the total number of the features detected depends on threshold constraints, while threshold is in a high value, improved SURF with HE still can obtain a large number of features, which leads to more matched feature pairs for accurate estimation of the object position. The number of the matched features obtained by improved SURF with HE is at least 2.2 times more than that obtained without HE (see Table 3).

4.3. Automatic image stitching with our algorithm

Our stitching method is applied to microscope images (see Fig. 13). During the stitching process, the conventional method, such as Autostitch, is to match all features of the two images to be stitched, so the process is time-consuming. In our algorithm, phase correlation is employed to estimate the rough overlapping areas, then extract features in the overlapping zones and match the corresponding features. It is significant to reduce the amount

Table 4
Comparison between our method and the software for image stitching.

	Autostitch	Proposed methods
User input	No need	No need
Input format	JPEG	All common image formats
Preprocessing	No	Histogram equalization
Main algorithm	SIFT	Improved SURF
Translation	Yes	Yes
Zoom + rotation, blur, illumination, viewpoint	Partially invariant	Very robust
Change in exposure, uneven illumination	Partially invariant	Partially invariant
Field uneven and distortion	No	No, but can partially registration
Blending	Liner or Multi-Band	Liner
Stitching seam	No	No
Average time needed	4.2 s	2 s

of calculation and enhance the matching accuracy between the two microscope images to be stitched.

The comparison between common stitching software, Autostitch, and our method is listed in Table 4. The results show that neither our method nor the Autostitch need any user input. However, the input images are limited to JPEG by using Autostitch, while our method does not have any detailed requirements about the formats of input images (it is applicable to the image formats such as JPEG, GIF, PNG, TIFF, BMP, and PPM). The microscope images are preprocessed by HE in this study to increase the number of features and the algorithm proposed in this study is more robust to zoom, rotation, blur, illumination and viewpoint than SIFT and SURF. The limitation is that only part of the features can be matched with changes of exposure, uneven illumination, field uneven. The main purpose of applying linear fusion algorithm is to reduce the computation amount and the experiment results demonstrate that this algorithm can meet the requirement of seamless image stitching. The proposed algorithm is faster than Autostitch and the averaged time consumed by our method is only half of that by the software. The stitching result demonstrates that our method can be applied to stitching microscope images seamlessly.

Due to the limitation of experiment condition, the stitching experiment has just been conducted for images from normal light microscope in this study, not for images from other microscopes, such as electron microscope and fluorescence microscope. Autostitch software is applicable to images from fluorescence microscope and electron microscope (Ma et al., 2007b). The main algorithm of Autostitch is SIFT, while SURF is just a speeded up one based on SIFT. Therefore, it can be predicted that the improved SURF is also applicable to images from electron microscope and the method in this paper may also be applicable to images from fluorescence microscope.

5. Conclusion

This paper proposed a method for fast automated microscope image stitching, in which the accurate registration for images with great changes of viewpoint is accomplished by using the improved SURF to simulate various image affine transformations. Our Experiments demonstrate that the improved SURF is more robust to changes of viewpoint, rotation, zoom, blur and illumination than SURF. The proposed stitching method is applicable to various formats of images from normal light microscope, and is faster than the stitching software. In the future, we plan to write a software based on this method to provide applications of image stitching for biological researches.

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